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WORKING EXPERIENCE

➢ **Thomson Reuters**, *London, United Kingdom* *April 2026 - Present*
***Applied Scientist Intern**, TR Lab*
Team: CoCounsel Drafting Team & CoCoNext Team (Legal Agent and CoCoBench Eval)
➢ *Contributions* - ***(i)*** **[Contribution to *AutoEval*]** Architected an end-to-end multi-turn evaluation platform, using graph-based generation to produce multiple quality-gated scenarios and integrating stateful user simulation with automated rollout judges to scale legal-drafting agent testing. ***(ii)*** **[Contribution to *CoreAgent*]** Engineered secure and high-fidelity DOCX ↔ LegalMarkdown conversion that preserves complex formatting and revisions, validated across $15K$ legal documents. ***(iii)*** **[Contribution to *CoCoBench*]** Built and calibrated the legal-document redline LLMaaJ evaluation pipeline with tracked-change parsing, multiple-run consensus, and automated testing across >100 expert-annotated cases, improving agreement 10 percentage points. The regression test is automated weekly for CoCoNext Agent.

➢ **Imperial College London**, *London, United Kingdom* *October 2025 - Present*
***Postdoc**, Intelligent Digital Systems Lab (iDSL)* *PI: Prof. Christos Bouganis*
Description: Optimisation and mapping of generative AI models on GPUs and on the cloud

EDUCATION

➢ **University of Liverpool**, *Liverpool, United Kingdom* *October 2021 - September 2025*
***Ph.D.** in Computer Science* *Supervisors: Prof. Xiaowei Huang, Prof. Xinping Yi*
Thesis: Advancing Geometric Deep Learning: Deep Architectures and Latent Geometry [PhD Thesis]
➢ **Australian National University**, *Canberra, Australia* *July 2019 - June 2021*
***Bachelor** of Advanced Computing (Hons)* *Supervisors: Prof. Tom Gedeon, Dr. Zhenyue Qin*
Thesis: A Remedy for Negative Monte Carlo Estimated Values of KL-Divergence
➢ **Shandong University**, *Shandong, China* *July 2017 - June 2019*
***Bachelor** of Computer Science and Engineering*

ABOUT ME

I am a Postdoctoral Research Associate in the Intelligent Digital Systems Lab (iDSL), Imperial College London, working with Prof. Christos Bouganis and his team. I am also a Applied Scientist in Thomson Reuters, working on CoCoNext Agent & its evaluation. Specifically, my work focus on the following topics:

- ➢ Efficient AI (*e.g.,* Hardware-Aligned Linear/Quasilinear Scalable Transformers) [TMLR'25], [TNNLS'26]
- ➢ Geometric Deep Learning (*e.g.,* (Non-Euclidean) Graph Embedding) [TAI'24], [ECML'24]
- ➢ Human-AI Preference Alignment (*e.g.,* RLHF, AISafty) [TechReport'24]
- ➢ Agentic System and Evaluation

I hold the **Global Talent Visa (2025 - 2030), which allows me to work in the UK without requiring visa sponsorship.*

ACADEMIA PROJECT EXPERIENCE

HAMLET: Human-centered generative Ai fraMework for culturaL industriEs' digital Transition
***Postdoc** with iDSL, Imperial College London* *2025 - Present*
➢ *Project Page* - [https://hamlet-project.eu/] (Funder: EU Horizon Programme)
➢ *Objective* - ***(i) Productionised LLM/DiT inference optimization for consortium models***, targeting latency/throughput and resource constraints. ***(ii) Engineered an end-to-end model optimization and serving pipeline*** by leveraging NVIDIA ModelOpt for post-training optimization (*e.g.* real quantization (W4A16(GPTQ)/W8A8(SQ)/NVFP4), network pruning, and sparsification). Deployed these optimized models within a Customized Inference Server Docker with a modified vLLM-Omni

serving backend, achieving production-grade throughput and energy efficiency. ***(iii) Developed linear-time attention architecture (GatedFWA)*** (Triton kernel fusion, FlashAttention-compatible sliding-window attention extension) to reduce IO overhead and improve long-context performance of causal Attention. GatedFWA delivers competitive throughput with negligible overhead and better use of global context on autoregressive LLM **Pre-Training→Mid-Training→SFT→RL** benchmarks.
≻ *Output* - Supported jointly by HAMLET (EU Horizon Grant 101178362) and GENCI–IDRIS (Grant 2025-AD011014447), we published paper that innovates Transformer attention computation through gating Associative Memory [IEEE TNNLS'26].

EnnCore: End-to-End Conceptual Guarding of Neural Architectures
Research Assistant *with TACPS Lab, University of Liverpool* *2023 - 2024*
≻ *Project Page* - [https://enncore.github.io/] (Funder: EPSRC)
≻ *Objective* - ***(i) Developed conceptual guardrails for LLMs*** to ensure robustness against adversarial attacks. Engineered a *Sentinel* prefix-model that intercepts and refines toxic prompts via a adversarial training framework, optimizing defense strategies using Multi-Agent PPO, which aligns with the project-goal of safeguarding conversational agents against unexpected behaviors. ***(ii) Enhanced the structural integrity and scalability of neural architectures*** by resolving information bottlenecks in long-sequence modeling. Designed a linear-complexity attention mechanism utilizing compressed token propagation to maintain global context without the quadratic cost of standard Transformers.
≻ *Output* - Supported by this project (EPSRC Grant EP/T026995/1), we published several papers on efficient Transformers with compressive memory [TMLR'25] and human-AI preference alignment (RLHF) of modern LLMs [TechReport'24].

Advancing Geometric Deep Learning: Deep Architectures and Latent Geometry
Ph.D. *research with the University of Liverpool* *2021 - 2025*
≻ *Objective* - ***(i) Overcame depth limitations in Hyperbolic GNNs*** by formulating a hyperbolic Dirichlet energy metric to diagnose over-smoothing. Based on GeoOpt framework, we engineered a scalable hyperbolic feature transformation layer and residual framework, enabling the first effective training of deep hyperbolic networks that capture high-order hierarchical structures. ***(ii) Developed continuous geometric diffusion models*** by reframing graph propagation as a Hyperbolic Neural PDE. Derived novel numerical integrators (*e.g.,* Hyperbolic Runge-Kutta) and an anisotropic diffusivity mechanism to model complex local-global dependencies on Riemannian manifolds with constant parameter efficiency.
≻ *Output* - We published paper about state-of-the-art deep (non-Euclidean) GNN architecture [IEEE TAI'24] and its constant backprop memory augmentation [ECML'24].

ZeroDriver: Local-Deployable Safety-Aware LLM-Enabled Trajectory Planner for Autonomous Driving
Research Assistant *with Institute of Software, Chinese Academy of Sciences* *2024 - 2024*
≻ *Project Page* - [https://ljxw88.github.io/zerodriver/] (Funder: Key Laboratory of System Software, ISCAS)
≻ *Objective* - ***(i) Engineered a multi-LLM-agent trajectory planning framework*** utilizing Vision LLMs (Qwen-VL) to enhance autonomous decision-making and safety in complex environments. ***(ii) Designed a modular architecture*** integrating perception, memory, and reasoning components to optimize path planning. ***(iii) Validated system efficacy*** via open-loop benchmarks on the nuScenes, demonstrating robust performance for autonomous driving and low-altitude economy applications.
≻ *Output* - This research is published in NeurIPS as a workshop paper [SafeGenAi'24].

OPEN SOURCE CONTRIBUTION

Flash Linear Attention (FLA)
Contributor - [https://github.com/fla-org/flash-linear-attention] *5k+ GitHub Stars*
≻ *Repo* - A widely-adopted open-source library providing hardware-efficient Triton kernels for linear attention variants (*e.g.,* GatedDeltaNet, Mamba2, RetNet, RWKV), a standard backbone for linear-complexity LLM research and pre-training.
≻ ***Identified and resolved critical model-loading bugs*** affecting three linear attention architectures (GatedDeltaNet, Comba, KimiDeltaAttention) under `transformers`≥ 5.0, where loading pre-trained checkpoints silently produced corrupted outputs. Delivered an idempotent fix across all three model files and provided a migration utility for existing checkpoint users.

herdrm
Contributor - [https://github.com/missuo/herdrm] *∼1k GitHub Stars*
≻ *Repo* - Native macOS console for herdr, which enables seeing Claude Code, Codex, Gemini, Grok and OpenCode across your Mac and every SSH box you own, and jump into any one of them at full-TUI fidelity.
≻ ***Currently the second biggest contributor***. Contributed remote SSH support (Keychain-backed auth, Tailscale), cross-agent file/image paste with secure upload, and terminal UX fixes to herdr (open-source macOS coding-agent console), landing multiple merged PRs upstream.

TEACHING

Teaching Assistant EEECS, University of Liverpool

- COMP338 (Computer Vision) *2023 - 2024*
- ELEC230 (Robotics) *2021 - 2022*
- COMP534 (Applied AI) *2021 - 2022*

Academic Tutor CECS, Australian National University

- COMP3600/6466 (Algorithms) *2020 - 2021*

SERVICES

Conference Reviewer

- Program Committee, ICLR, 2025, 2026
- Program Committee, NeurIPS, 2024, 2025
- Program Committee, AAAI, 2025, 2026

Journal Reviewer

- Invited Reviewer, Frontiers in Neurorobotics
- Invited Reviewer, Frontiers in Public Health

MISCELLANY

- **Pure Data Loop Pedal (Electromusic from Signal Processing)** *2020*

A pure data based patch, cross-platform performance interface specially designed for Novation Launchkey Mini MK3. This is also a Laptop Ensemble artefact created as part of the LENS ANU Laptop Ensemble '21 cohort.
GitHub - [https://github.com/ljxw88/pure-data-loop-pedal]
Live Performance (YouTube) - [https://youtu.be/A0NG-T8y7gM]
Live Performance (Bilibili) - [https://www.bilibili.com/video/BV1c54y1H7uX?share_source=copy_web]

- CS Outreach, University of Liverpool *2021 - 2022*
- Guild Associate Representative, School of Mechanical, Electrical & Information Engineering, SDU *2018 - 2019*
- Shandong University (at Weihai), ACM-ICPC School Team Member *2018 - 2019*

PUBLICATIONS

In Proceedings (Journal)

GatedFWA: Linear Flash Windowed Attention with Gated Associative Memory *2026*
≻ ***Jiaxu Liu***, *Yuhe Bai, Xiangyu Yin, Christos-Savvas Bouganis*
◦ IEEE Transactions on Neural Networks and Learning Systems (**IEEE TNNLS**)
◦ Publication - [https://www.arxiv.org/abs/2512.07782]

Toward Linearly Regularizing the Geometric Bottleneck of Linear Generalized Attention *2025*
≻ ***Jiaxu Liu***, *Xinping Yi, Xiangyu Yin, Yuhang Song, Gaojie Jin, Xiaowei Huang*
◦ Transactions on Machine Learning Research, *PMLR* (**TMLR**)
◦ Publication - [https://openreview.net/forum?id=Vpyg3fqXbl]
◦ Code - [https://github.com/ljxw88/Regularizing-Geometric-Bottleneck]

DeepHGCN: Toward Deeper Hyperbolic Graph Convolutional Networks *2024*
≻ ***Jiaxu Liu***, *Xinping Yi, Xiaowei Huang*
◦ IEEE Transactions on Artificial Intelligence (**IEEE TAI**)
◦ Publication - [https://ieeexplore.ieee.org/abstract/document/10632071]
◦ Code - [https://github.com/ljxw88/deephgcn]

In Proceedings (Conference)

Dynamic Context Adapters: Efficiently Infusing History into Vision-and-Language Models *2026*
≻ *Yuhang Song, Bor-Jiun Lin,* ***Jiaxu Liu***, *Te-Chuan Chiu, Anh Nguyen, Chun-Yi Lee*
◦ International Conference on Pattern Recognition (**ICPR 2026**)
◦ Publication - [https://link.springer.com/chapter/10.1007/978-3-032-31663-9_8]

Enhancing Robust Fairness via Confusional Spectral Regularization *2025*
≻ *Gaojie Jin, Sihao Wu,* ***Jiaxu Liu***, *Tianjin Huang, Ronghui Mu*
◦ International Conference on Learning Representation (**ICLR 2025**)
◦ Publication - [https://openreview.net/forum?id=lW0ZndAimF]

Continuous Geometry-Aware Graph Diffusion via Hyperbolic Neural PDE *2024*
≻ ***Jiaxu Liu***, *Xinping Yi, Sihao Wu, Xiangyu Yin, Tianle Zhang, Xiaowei Huang, Shi Jin*
◦ European Conference on Machine Learning (**ECML 2024**)
◦ Publication - [https://link.springer.com/chapter/10.1007/978-3-031-70352-2_14]
◦ Code - [https://github.com/ljxw88/HyperbolicGD]

PRASS: Probabilistic Risk-averse Robust Learning with Stochastic Search *2024*
≻ *Tianle Zhang, Yanghao Zhang, Ronghui Mu,* ***Jiaxu Liu***, *Jonathan Fieldsend, Wenjie Ruan*
◦ International Joint Conferences on Artificial Intelligence (**IJCAI 2024**)
◦ Publication - [https://www.ijcai.org/proceedings/2024/62]

DeepGRE: Global Robustness Evaluation of Deep Neural Networks *2024*
≻ *Tianle Zhang,* ***Jiaxu Liu***, *Yanghao Zhang, Ronghui Mu, Wenjie Ruan*
◦ IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP 2024**)
◦ Publication - [https://ieeexplore.ieee.org/abstract/document/10446884]
◦ Code - [https://github.com/TrustAI/DeepGRE]

Representation-Based Robustness in Goal-Conditioned Reinforcement Learning *2024*
≻ *Xiangyu Yin, Sihao Wu,* ***Jiaxu Liu***, *Meng Fang, Xingyu Zhao, Xiaowei Huang, Wenjie Ruan*
◦ Annual AAAI Conference on Artificial Intelligence (**AAAI 2024**)
◦ Publication - [https://ojs.aaai.org/index.php/AAAI/article/view/30176]
◦ Code - [https://github.com/TrustAI/ReRoGCRL]

Declarative Residual Network For Robust Facial Expression Recognition *2020*
≻ *Ruikai Cui, Josephine Plested,* ***Jiaxu Liu***

○ International Conference on Neural Information Processing (**ICONIP 2020**)
○ Publication - [https://link.springer.com/chapter/10.1007/978-3-030-63820-7_39]
○ Code - [https://github.com/CuiRuikai/Bidirectional-Residual-Declarative-Network]

WORKSHOPS & PREPRINTS

Workshops

Integrating Object Detection Modality into VLM for Enhanced Autonomous Driving Agent *2024*
≻ *Lingfeng He, Yiming Sun, Sihao Wu, **Jiaxu Liu (project lead)**, Xiaowei Huang*
○ NeurIPS 2024 Workshop SafeGenAi
○ Publication - [https://openreview.net/pdf?id=v8fYRyerUS]

FacePhi: Lightweight Multimodal Large Language Model for Facial Landmark Emotion Recognition *2024*
≻ *Hongjin Zhao, Zheyuan Liu, Yang Liu, Zhenyue Qin, **Jiaxu Liu**, Tom Gedeon*
○ ICLR 2024 Workshop PML4LRS
○ Publication - [https://openreview.net/pdf?id=7Tf9JTGeLU]

Preprints

Crossing the Margin Cliff: Toward Relearn-Robust LLM Unlearning via Margin Calibration *2026*
≻ *Xiangyu Yin, **Jiaxu Liu (equal)**, Zhen Chen, Chih-Hong Cheng*
○ Preprint - [https://arxiv.org/abs/2607.27836]

CeTAD: Towards Certified Toxicity-Aware Distance in Vision Language Models *2025*
≻ *Xiangyu Yin, **Jiaxu Liu (equal)**, Zhen Chen, Jinwei Hu, Yi Dong, Xiaowei Huang, Wenjie Ruan*
○ Preprint - [https://arxiv.org/abs/2503.10661]

Robust RL with LLM-Driven Data Synthesis and Policy Adaptation for Autonomous Driving *2024*
≻ *Sihao Wu, **Jiaxu Liu (equal)**, Xiangyu Yin, Guangliang Cheng, Xingyu Zhao, Meng Fang, X. Yi, X. Huang*
○ Preprint - [https://arxiv.org/abs/2410.12568]

Tiny Refinements Elicit Resilience: Toward Efficient Prefix-Model Against LLM Red-Teaming *2024*
≻ ***Jiaxu Liu**, Xiangyu Yin, Sihao Wu, Jianhong Wang, Meng Fang, Xinping Yi, Xiaowei Huang*
○ Preprint - [https://arxiv.org/abs/2405.12604]

Curvature-Aware Graph Neural Diffusion: Hyperbolic Vector Flow and Numerical Integrator *2024*
≻ ***Jiaxu Liu**, Xinping Yi, Sihao Wu, Xiangyu Yin, Tianle Zhang, Xiaowei Huang, Shi Jin*
○ Under Review

A Remedy for Negative Monte Carlo Estimated Values of KL-Divergence *2021*
≻ *Zhenyue Qin, **Jiaxu Liu**, Yang Liu, Saeed Anwar, Pan Ji, Tom Gedeon*
○ Code - [https://github.com/ljxw88/pvae-tangent-estimator]
○ Thesis & Preprint - [https://github.com/ljxw88/pvae-tangent-estimator/blob/master/data]